

Sowhardo Bin Atik

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[GitHub](#) · [Portfolio](#) · [LinkedIn](#)

Objective

Seeking a **Backend-Focused Software Development** role. Experienced in building **RESTful APIs** with **Node.js (Express.js NestJS)**, designing database schemas in **PostgreSQL**, integrating **AI**, and implementing caching with **Redis** and background job processing with **BullMQ**.

Education

BRAC University	2022 - June 2026
Bachelor of Science in Computer Science and Engineering	CGPA: 3.70 / 4.00

Technical Skills

Languages:	JavaScript, TypeScript, Python, Java, C, C++
Backend & APIs:	Node.js, Express.js, NestJS, REST APIs, Python (basic FastAPI/Django)
Frontend:	HTML, CSS, Next.js, React.js, Tailwind CSS
Databases:	SQL, PostgreSQL, pgvector, Redis, Prisma ORM, Indexing, Schema Design
Security & Auth:	JWT Authentication, Role-Based Authorization, Data Validation, Helmet, CORS
DevOps & Tools:	Git, GitHub, Docker, Linux, Cloud Computing (Basics)
Network:	HTTP/HTTPS, TLS/SSL, DNS, CDN, Subnetting, Proxy
Testing:	Jest, Playwright, Postman
Problem Solving:	LeetCode [Profile]

Projects

MetroSheba: AI-Driven Transit & Ticketing Platform	[Live] · [Video] · [GitHub]
<i>Technologies: Express.js, Node.js, React, PostgreSQL, Redis, RAG</i>	

- Developed a full-stack ticketing platform with **RESTful backend APIs** featuring secure session handling, user accounts, and an automated transaction routing workflow.
- Programmed an integrated RAG-based search API engine backed by semantic vector embeddings (**pgvector**) to successfully resolve dynamic client routing queries.
- Optimized data retrieval pipelines by configuring a **Redis caching layer**, minimizing recurrent primary relational database requests and cutting API overhead.
- Assured structural code health by constructing backend unit tests with **Jest** and end-to-end tests with **Playwright**, deploying code health gates via a continuous integration pipeline using **GitHub Actions**.

Field Dispatch System	[GitHub]
<i>Technologies: Next.js, TypeScript, Tailwind CSS, NestJS, PostgreSQL, MongoDB, Redis, Socket.IO, Bull Queue</i>	

- Built a responsive **Field Service Management** frontend with **Next.js, TypeScript**, and **Tailwind CSS**, featuring JWT auth, Zustand state management, and role-based dashboards.
- Designed an **AI-driven technician allocation algorithm** that classifies job type, scores technicians by **Haversine distance** and active workload, and dynamically ranks candidates within a row-locked **PostgreSQL** transaction to prevent race conditions.
- Architected a scalable **NestJS** backend offloading assignment logic to a **Bull Queue** with **Redis** for non-blocking, concurrent processing with automatic retries, plus real-time updates via **WebSockets (Socket.IO)**.
- Secured the system with **JWT token rotation**, **RBAC** across Client, Technician, Dispatcher, and Admin roles, and Redis-backed token blacklisting for session revocation.

Achievements

AI CERTs Prompt Engineer Level 1 Certification	2024
Learned LLM architecture fundamentals to apply advanced prompting methodologies (zero/few-shot, Chain-of-Thought, and Role prompting).	
Anthropic Certifications	2026
Completed Claude Code 101, Claude Code in Action, Introduction to Agent Skills, Introduction to Subagents, Introduction to Model Context Protocol, and AI Fluency for Builders.	
[View All Certificates]	